

Nicolas F. Nino

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WORK & LEADERSHIP EXPERIENCE

Air Force Research Laboratory

Research Engineer Intern (Secret Security Clearance)

Eglin AFB, FL

May 2025 - Aug 2025

- **Trained** and optimized autonomous agents using Proximal Policy Optimization (PPO) in Stable Baselines for complex aerial navigation tasks in dynamic environments.
- **Engineered** advanced control systems, including geometric and other nonlinear controllers, within a custom high-fidelity simulation environment.
- **Leveraged** in-depth physics knowledge to drive innovation, enhancing simulation realism and reducing the margin of error in mission-critical test environments.

Air Force Research Laboratory

Research Engineer Intern (Secret Security Clearance)

Eglin AFB, FL

Aug. 2024 - May 2025

- **Re-architected** full drone simulation in UE5, implementing a cascaded PID controller for rate, angle, velocity, and position control.
- **Built** a full ROS2 amd ZeroMQ communication network within UE5, supporting real-time data exchange and control interfaces.
- **Trained** a reinforcement learning agent using PPO in Unreal Engine to reach target positions and avoid obstacles in a dynamic 3D environment.
- **Developed** dynamic UI tools and added gamepad integration for manual and semi-autonomous control testing.

Air Force Research Laboratory

Research Engineer Intern (Interim Security Clearance)

Eglin AFB, FL

May 2024 - Aug 2024

- **Designed** and built a physics-accurate quadrotor dynamics model in Unreal Engine using C++ and blueprint scripting.
- **Developed** initial PID controller for position control and created a modular drone framework to support future expansion.
- **Conducted** performance testing to compare simulation accuracy with real-world drone logs.

Education

University of Central Florida

Bachelor of Science in Data Science

Orlando, FL

May 2021 - Present

Relevant Coursework

Mathematics

- Linear Algebra
- Advanced Calculus I, II
- Probability and Statistics

Algorithms & Theory

- Data Structures & Algorithms I, II
- Logic and Proof
- Discrete Structures

AI & Robotics

- Robot Vision
- AI in Video Game Programming
- Math Foundations for ML

Technical Skills

Programming Languages

- C, C# , C++
- Python, Java
- R, Rust
- Matlab, Assembly

Simulation & Modeling

- Unreal Engine 5
- Isaac Sim, Gazebo
- Webots, Simulink
- Blender
- UE Blueprints

Control & ML Systems

- ROS 1 & 2, PX4
- ArduPilot, ZeroMQ
- HITL, SITL, PID
- PPO, SB3, Gym
- Real-Time Systems

Research Interests

Multi-Agent Systems, Adaptive Control, Reinforcement Learning, Geometric Mechanics, Game Theory, Graph Theory, Robotics, Simulation